CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Second & Fourth Semester of BSc Physics Examination May 2018
(Elective Course)
PD156 Statistics-II

Date:03/05/2018,Thursday **Time:01:30 p.m. to 02:00 p.m.** **Maximum Marks:20**

MCQ

Important Instructions:

- Tick the correct answer/s and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

1. Let A and B be any two events of sample space S, Then the expression for atleast one of the events A and B occur is______.

(i) $A \cup B$ **(iii)** $A \cap B'$

(ii) $A \cap B$ **(iv)** $A' \cap B$

2. If $P(A \cup B) = 0.64, P(A \cap B) = 0.16$ and A and B are independent events then $P(A) =$______.

(i) 0.8 **(iii)** 0.25

(ii) 0.48 **(iv)** 0.4

3. If $P(A) = 0.9$,$P(B) = 0.8$ and $P(A \cap B) = 0.75$ then $P(A \cap B') =$______.

(i) 0.95 **(iii)** 0.18

(ii) 0.15 **(iv)** 0.1

4. If A_1,A_2 and B are events of a sample space $\mathcal{S}$, then $P(A_1 \cup A_2|B) =$ ______.

(i) $P(A_1|B) + P(A_2|B)$

(ii) $P(A_1|B) - P(A_1 \cap A_2|B)$

(iii) $P(A_1|B) + P(A_2|B) - P(A_1 \cap A_2|B)$

(iv) $P(A_1|B)P(A_2|B)$

5. Three events A, B and C are pairwise independent if ______.

(i) $P(A \cap B) = P(A)P(B)$

(ii) $P(A \cap C) = P(A)P(C)$

(iii) $P(C \cap B) = P(C)P(B)$

(iv) All **(i)**,**(ii)** and **(iii)**

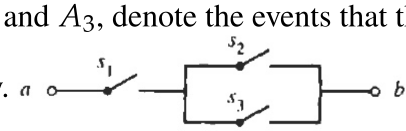
6. If $P(A) = 0.9$ and $P(B) = 0.8$. Then $P(A \cap B) \geq$ ______.

(i) 0.3

(ii) 0.7

(iii) 0.1

(iv) 0.2

7. Let A_1, A_2 and A_3, denote the events that the switches s_1, s_2 and s_3 are closed, respectively. 

. Let A_{ab}, denote the event that there is a closed path between terminals a and b. Then $A_{ab} =$______.

(i) $A_1 \cup A_2 \cup A_3$

(ii) $(A_1 \cap A_2) \cup A_3$

(iii) $A_1 \cap (A_2 \cup A_3)$

(iv) $A_1 \cap A_2 \cap A_3$

8. If X is the r.v. whose cdf is given by $F(x) = \begin{cases} 0 & x < 0 \\ x + \frac{1}{2} & 0 \leq x < \frac{1}{2} \\ 1 & x \geq \frac{1}{2} \end{cases}$.

Then $P\left(X \leq \frac{1}{4}\right) =$____________

(i) $\frac{3}{4}$

(ii) $\frac{1}{2}$

(iii) $\frac{1}{4}$

(iv) 0

9. Let X be a continuous r.v. X with pdf $f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{Otherwise} \end{cases}$.

Then $P(0 < X \leq 2) =$ ____________.

(i) $\frac{1}{4}$

(ii) 1

(iii) 0

(iv) 4

10. If $M_X(t)$ is a moment generating function of a random variable X.
Then $E(X) =$______.

(i) $M'_X(0)$

(ii) $M''_X(0)$

(iii) $M''_X(t)$

(iv) $M'_X(t)$

11. The moment generating function of a random variable X with pdf $f(x)$, is $\frac{e^t - 1}{t}, t \neq 0$. Then $E(X) =$______.

(i) 0

(ii) 1

(iii) $\frac{1}{2}$

(iv) 2

12. If the probability density function of Beta distributed random variable X is

$$f(x) = \begin{cases} \frac{1}{12}x^2(1-x), & 0 < x < 1 \\ 0, & \text{Otherwise} \end{cases}$$

Then $E(X) =$______.

(i) $\frac{2}{5}$

(ii) $\frac{5}{3}$

(iii) $\frac{5}{2}$

(iv) $\frac{3}{5}$

13. If random variable X has pdf $f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b; a, b \in \mathcal{R} \\ 0, & \text{Otherwise} \end{cases}$,

Then $P\{|X - E(X)| \leq k\sqrt{V(X)}\} =$ ______.

(i) $\frac{k}{\sqrt{3}}$

(ii) $\frac{k}{3}$

(iii) $\frac{\sqrt{3}}{k}$

(iv) $\frac{\sqrt{3}}{k}$

14. If X is a random variable such that $E(X) = 3, V(X) = 4$,
then $P(-2 < X < 8) \geq$ ______.

(i) 0.16

(ii) 0.4

(iii) 0.84

(iv) 0.6

15. The Moment generating function of distribution $P(X = x) = \begin{cases} p, & x = 1 \\ q, & x = 0 \\ 0, & \text{Otherwise} \end{cases}$

is ______.

(i) $\dfrac{1}{q + pe^t}$

(ii) $q + pe^t$

(iii) $p + qe^t$

(iv) $\dfrac{1}{p + qe^t}$

16. For Exponential distribution of a random variable X which is true? ______.

(i) $E(X) = [V(X)]^2$

(ii) $[E(X)]^2 = V(X)$

(iii) $E(X) = V(X)$

(iv) $E(X) = \sqrt{V(X)}$

17. If the m.g.f of random variable X is $M_X(t) = (\frac{1}{3} + \frac{2}{3}e^t)^5$,
then $P(X = 5) =$ ______.

(i) $\left(\dfrac{2}{3}\right)^5$

(ii) $\left(\dfrac{1}{3}\right)^5$

(iii) $\dfrac{2}{3}$

(iv) $\dfrac{1}{3}$

18. If random variable X has Poisson distribution with parameter μ, then Moment generating function $M_X(t) =$______.

(i) $e^{\mu(e^t - 1)}$

(ii) $e^{\mu}(e^t - 1)$

(iii) $e^{-\mu(e^t - 1)}$

(iv) $e^{-\mu}(e^t - 1)$

19. Let X be the number of heads (successes) in $n = 7$ independent tosses of an unbiased coin. Then $M_X(t) =$______.

(i) $(1 + e^t)^7$

(ii) $\dfrac{1}{128}(1 + e^{-t})^7$

(iii) $\dfrac{1}{128}(1 + e^t)^7$

(iv) $(1 + e^{-t})^7$

20. The cumulative distribution function of pdf

$f(x) = \begin{cases} \dfrac{1}{b - a}, & a < x < b \\ 0, & \text{Otherwise} \end{cases}, a, b \in \mathcal{R}$ is______.

(i) $\dfrac{x - b}{b - a}$

(ii) $\dfrac{x - a}{b - a}$

(iii) $\dfrac{b - a}{x - a}$

(iv) $\dfrac{1}{b - a}$

End of **MCQ**

Candidate seat no:________________

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Date:03/05/2018,Thursday **Time:02:00 p.m. to 04:30 p.m.** **Maximum Marks:50**

Instructions:

- **Section I and II must be attempted in TWO ANSWER SHEET**.
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION I

Q - II Answer the following questions as directed. **[20]**

1. Define the following terms. **[08]**

(a) Axiomatic definition of Probability.

(b) Baye's Theorem (Statement Only).

(c) Probability density function.

(d) Probabilit mass function.

2. Attempt any three questions. **[12]**

(i) An urn contains five balls numbered 1 to 5 of which the first three are black and the last two are gold. A sample of size 2 is drawn without replacement. Let B_1 denote the event that the first ball drawn is black and B_2 denote the event that the second ball drawn is black.

(a) Describe a sample space for the experiment, and exhibit the events B_1,B_2 and $B_1 \cap B_2$.

(b) Find $P[B_1], P[B_2]$ and $P[B_1 \cap B_2]$.

(ii) Consider the experiment of tossing two dice. Assume that each sample point is equally likely. Define the events A, B, and C:

A = "First die results in a 1, 2, or 3."
B = "First die results in a 3, 4, or 5."
C = "The sum of the two faces is 9."

Determine whether the events A, B and C are independent or pairwise independent.

(iii) If X is the number of white balls in a sample of size 2 drawn without replacement from an urn containing 6 balls of which 4 are white, then determine (a) all possible values of X (b) Probability distribution of X (c) $E(X)$

(iv) Determine the mean and variance of probability distribution

$$P(X = x) = \begin{cases} e^{-\lambda}\dfrac{\lambda^x}{x!}, \lambda > 0, & x = 0, 1, 2, \ldots \\ 0, & \text{Otherwise} \end{cases}$$

(Do not use moment generating function)

SECTION II

Q - III Answer the following questions. **[30]**

1. Explain in brief **[12]**

(i) Chebyshev's Inequality.

(ii) Weak law of large numbers.

(iii) Lindeberg-Lévy Central Limit Theorem.

2. Attempt any three questions. **[18]**

(i) Show that the moment-generating function of the random variable X having a Normal probability distribution with mean μ and variance σ^2 is given by $M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$

(ii) From a lot of 10 missiles, 4 are selected at random and fired. If the lot contains 3 defective missiles that will not fire, what is the probability that **(a)** all 4 will fire? **(b)** at most 2 will not fire?

(iii) Buses arrive at a specified stop at 15- minutes intervals starting at 7:00 a.m. That is, they arrive at 7:00, 7: 15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7:00 and 7:30 find the probability that he waits **(a)** less than 5 minutes for a bus. **(b)** more than 10 minutes for a bus. **(c)** between 5 minutes and 10 minutes.

(iv) Obtain the moment generating function of Negative Binomial distribution with parameters r and p and hence find Mean and Variance of distribution.

(v) Impurities in batches of product of a chemical process reflect a serious problem. It is known that the proportion of impurities X in a batch has the Beta density function $f(x) = \begin{cases} 10(1-x)^9 & 0 \leq x \leq 1 \\ 0 & \text{Otherwise} \end{cases}$

(a) A batch is considered not sellable and then not acceptable if the percentage of impurities exceeds 60%. With the current quality of the process,what is the probability that a batch is not acceptable?

(b) Determine the mean proportion of impurities in the batch.

(c) Determine the standard deviation of proportion of impurities in the batch.

(vi) Suppose that the length of phone calls in minutes is an exponential random variable with parameter $\lambda = \dfrac{1}{10}$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait **(a)** more than 10 minutes; **(b)** between 10 and 20 minutes. **(c)** Determine the mean and standard deviation of length of phone call.